Evidence-Monotone Terrain Representation (EMTRF)

Supplementary Material

Table S1. Descriptive DTM support structure used to interpret site heterogeneity (referenced from main-text Section 5.3). Cell fractions are over each frozen crop. "M/I edge" is the fraction of horizontal/vertical adjacent supported-cell pairs with different measured/interpolated ancestry. "*M* dashed" is the share of measured cells whose support falls in the dashed band (33–66); main-text Table 2 instead counts measured-only contour segments drawn dashed, as a share of all segments. These are post-hoc mechanism diagnostics, not causal terrain effects.

Site	*M* cells	*I* cells	*U* cells	Median n_M	M/I edge	*M* dashed
White Sands	35.2%	32.8%	32.0%	12	0.94%	32.3%
Estonia Tava	76.3%	23.7%	0.0%	2	29.60%	50.6%
StREAM	66.9%	33.1%	0.0%	2	29.33%	59.8%
Marsh Island	0.8%	50.3%	48.9%	41	0.80%	15.6%

Table S2. Site-specific four-pipeline ablation and performance diagnostics at one source-cell tolerance (the per-site detail behind main-text Table 3). "EP promote" is the endpoint-only rate shared by P0/P1; "Grade loss" is the rate shared by P0/P2; "Recorded-EP loss" is P1 (recorded endpoint provenance). The P3 column reports support-promotion/provenance-loss counts. The EC-DP overhead per kilometre is a secondary microtiming diagnostic; the paired 5,305-vertex benchmark in the main text is the primary runtime estimate. Counts are the strict reconstruction, which retains every eligible span. The 2026 archived record contains 184 Marsh Island spans and 87.5% reduction, where the strict reconstruction yields 186; the two additional eligible records are one-edge feature runs. Because the original eligibility rule was not preserved, the strict reconstruction is reported here and the archived result is retained in the repository as a historical cross-check.

Site	Spans	Red.	EP promote	Grade loss	Recorded-EP loss	P3	+ms/km
White Sands	82	71.1%	24.4%	93.9%	0.0%	0/0	2.12
Estonia Tava	172	62.7%	6.4%	100%	7.6%	0/0	2.43
StREAM	54	67.4%	22.2%	100%	1.9%	0/0	2.05
Marsh Island	186	87.4%	4.3%	98.4%	0.0%	0/0	2.78

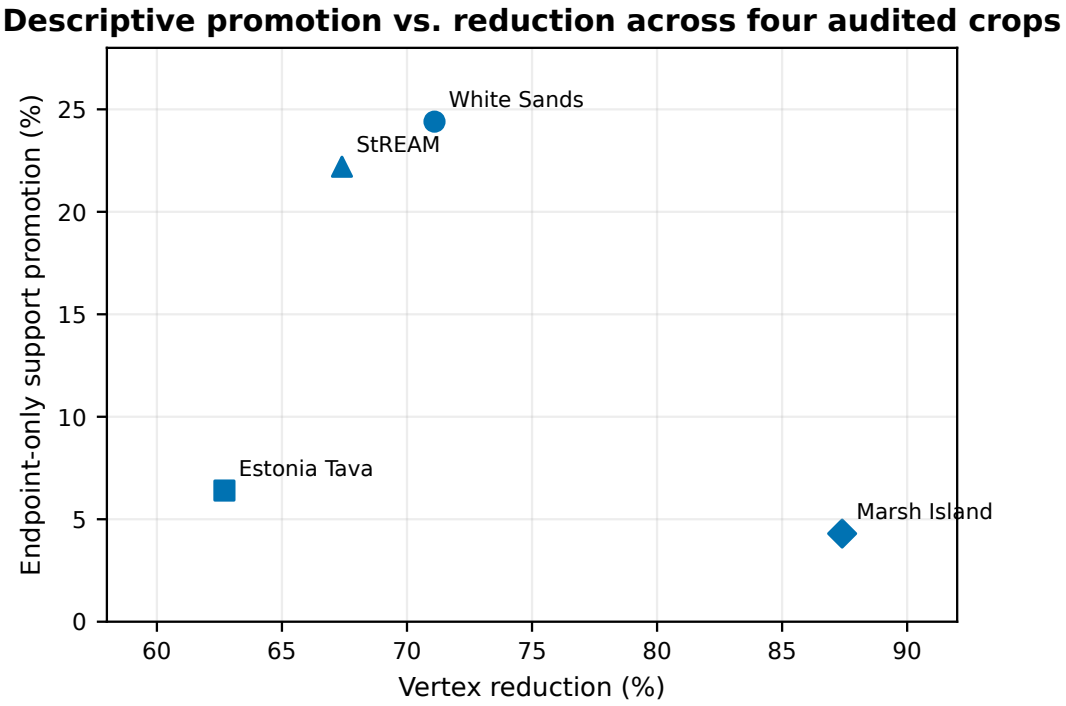


Figure S1. Descriptive endpoint-only support-promotion rate versus vertex reduction at one source-cell tolerance for the four audited field-derived crops. The points summarize heterogeneous crops and are shown only to expose the joint range of simplification and evidence failure. No fitted population relationship or causal association is claimed.

Supplementary Methods

S1. Adversarial source-arc fixtures. For each $\varepsilon \in \{0.02, 0.05, 0.10, 0.20, 0.40\}$, 2,000 deterministic 64-vertex polylines were generated from a low-amplitude sinusoidal geometry (10 units long, amplitude 0.01–0.19 units, so ε is in the same units) with a fixed mulberry32 seed family (seed `0xC0FFEE`+index). Source supports were sampled from $[0.78, 0.98)$; one to four non-endpoint vertices were then replaced by weak support in $[0.08, 0.50)$. Ordinary DP and EC-DP used the same geometric recursion. For every simplified span the endpoint-only value was the minimum of its retained endpoints, and the contract value was the minimum across the complete represented source arc; an endpoint-only promotion was counted when the former exceeded the latter, and a geometry mismatch when the retained index sets differed on fully supported fixtures. A separate gap experiment generated 1,000 fixtures, marked one interior vertex unsupported, and simplified at $\varepsilon = 0.25$; a baseline bridge was a DP segment whose retained endpoints lay on opposite sides of the unsupported vertex, whereas EC-DP split the source at that vertex so any crossing segment would violate the contract. A closed-ring extension used a deterministic 32-vertex ring with one weak interior value: when no semantic boundary supplies a start, the reference rule selects the lexicographically smallest (x, y) vertex, rotates the ring to that anchor, duplicates it to unroll the cycle, applies EC-DP, and re-closes the output, with complete-source evidence including the cyclic interval adjacent to the cut. All 32 cyclic reindexings were required to produce one canonical output signature.

S2. Real-data block-recovery audit. For each executed crop the study called the frozen production rasterization, confidence-aware DTM, and contour modules. Each emitted marching-squares segment lies within one known 2×2 DTM-cell block; the audit recovered that producing block geometrically and required its recomputed four-corner minimum confidence to match the emitted segment confidence. Source provenance was reconstructed from the four cell states, stored independently of visual grade: *M* when all four cells were measured-derived, *I* when all four were interpolated-derived, and *X* when both occurred. Unsupported corners cannot produce a segment because the contour extractor already refuses such cells.

S3. Standalone study components. The study is independent of the production application. It contains a JavaScript implementation of the reference support functions, directional classifier, ordinary DP, EC-DP, deterministic fixture generation, unsupported-gap injection, cyclic-ring handling, and timing harness. One independently written Python program checks the typed-evidence algebra, including the applicability, scope and lineage clauses: a channel that is not applicable to a source is ignored, whereas a channel applicable to a required source but carrying no value makes the output channel unavailable rather than computed from the remaining sources (a refusal preserved under composition), output scope is the intersection of the applicable sources' scopes, and lineage is the complete union over the source set. The same six clauses are also asserted against the shipped implementation by the exact conformance checks; a second Python program independently reimplements ordinary DP and EC-DP and is evaluated differentially on the same deterministic fixture coordinates and supports rather than by calling JavaScript output. Raw results, environment metadata, and checksums are frozen with the study artifacts.

S4. Timing protocol. For the synthetic benchmark, after five warm-ups ordinary DP and EC-DP were timed in 30 paired repetitions with alternating execution order, and retained-geometry checksums were required to match. The four-site real-contour benchmark used a site-normalized tolerance of one source grid cell (1.0 m for White Sands, Estonia Tava, and StREAM; 0.5 m for Marsh Island) over 289 polylines and 5,305 source vertices per pass; because one pass is sub-millisecond, each timed repetition batched 100 identical passes with five warm-ups and 30 alternating paired repetitions, then divided by 100. This is a post-contour representation-stage benchmark and excludes point loading, DTM construction, contour extraction, browser rendering, and GPU work. Median kernel time was 45.49 ms for DP and 59.03 ms for EC-DP on the 640,000 synthetic vertices (ratio 1.30), and the ratio was 1.032 on the four crops; Node.js timings on one machine.

S5. Excluded context crop. A USGS Coconino 3DEP context crop was executable but yielded only one stitched polyline and seven contour segments, too few for site-level quantitative comparison, and is therefore excluded from the quantitative corpus.

Supplementary source audit. The grade-to-label export mapping audited in the main text (`solid` → `measuredBacked`, `dashed` → `interpolatedBacked`) is present in the audited OpenLiDARViewer source. The canonical archive is the v0.6.5 release at git commit 4fb00a784d860a2e24cc1834d6c1230dac550355 (version DOI 10.5281/zenodo.21933671, source-archive SHA-256 fdb510e3416fb0cb02c71031bb7f653b8e0dde83ae9916c85d1c44e844b122fa). The identical mapping also appears unchanged in earlier v0.6.4 development snapshots (2026-08-09, SHA-256 71507a5115de8c8ad9125215e30222d87ca1cd400308fd464cebaac269373aaf and 1dfe94ef79838772730a04df66037b973a7be6b42abeecbf6f6ad7d6fd6c145f), confirming the counterexample is a stable export design carried across versions rather than a transient state of one archive.

Module reachability and test coverage. The shipped `simplifyPolyline` path pins vertices below the solid-confidence floor and vertices adjacent to grade transitions. The separate `contourAdaptiveGeneralize.ts` module likewise delegates geometry to per-feature DP, but its own source documentation states it has no production caller, so replacing that module alone would not make exported contour evidence conformant. In the updated test tree, evidence-registry and evidence-monotonicity tests reject authority upgrades in registered derivation chains and the E4 studies check selected numerical operators, whereas the adaptive-generalization tests exercise tolerance selection and geometry immutability/retention, not complete-source provenance/support serialization. Test reachability and operator conformance are therefore separate claims.

S6. Cross-implementation validation context (E4). The audited source defines E4 as agreement with a second implementation of the same computation, separate from E5 field ground truth. Two E4 records bear on the upstream geometry used in the main text: the mean 1 m DTM on the Estonia crop against a SciPy binned point-in-cell mean, and marching-squares isolines against GDAL on an analytic DEM. Both are author-run implementation-agreement cross-checks, not evidence that export provenance is correct, and not field-accuracy validation.

S7. Alternative methods. All methods receive the same input: the strict feature runs of the four crops (contours at one-cell intervals, joined, split at display-grade changes), with per-edge typed evidence, and the production DP distance rule at a one-cell tolerance. *DP*: ordinary DP; each span carries the evidence of its first and last source edge. *Length-weighted mean*: DP geometry; each span carries, per channel, the length-weighted mean support of the edges it replaces that have the channel, and the provenance union of those edges. *Boundary vertices*: a vertex is forced wherever the provenance class (M, I, X) changes between adjacent edges, and DP is applied between forced vertices with endpoint attribution. *Homogeneous splitting (b)*: the line is cut wherever provenance or typed support changes, with support compared after integer division by b points on the 0–100 scale ($b = 0$ means exact equality; $b = 10, 25$), and DP is applied to each run with endpoint attribution. *EC-DP*: DP geometry; each span carries the minimum typed support and provenance union of every edge it replaces. For every method a span is promoted when the support it carries exceeds the minimum over its replaced edges in either channel, and loses provenance when its carried provenance misses a class present in those edges. Length-weighted over- and understatement compare the span's carried support with each replaced edge's own value. Driver: `ablation/baselines.cjs`.

S8. Alternative support profiles. Each profile replaces the per-cell support field (0–100) before contour attribution; geometry is unchanged. Parameters were fixed before any run and not tuned. *Exponential*: measured cells $100(1 - e^{-n/2})$; interpolated cells $50\, e^{-(d-1)/1.5}$, with n the return count and d the distance to data in cells. *Kriging*: $100(1 - \sigma^2/s)$ with sill $s = 1$, where σ^2 is the simple-kriging (Gaussian-process posterior) variance at the cell from its $k = 8$ nearest measured cells within 6 cells, with exponential covariance (sill $s = 1$, range 3 cells) and observation noise τ^2/n_j, $\tau^2 = 0.5$; the computation was checked against an independent NumPy solution on six cells. *Standard error* (named SURVEY in the code; no survey checkpoints are used): $100 \exp(-\mathrm{se}^2/(2 \cdot 0.10^2))$, where se is the standard error of the cell elevation in metres: $\sqrt{(s_z^2 + 0.05^2)/n}$ for measured cells (s_z the sample standard deviation of the cell's returns, 0.05 m instrument precision) and $\sqrt{\mathrm{se}_{\mathrm{med}}^2 + (0.05\, d\, L)^2}$ for interpolated cells ($\mathrm{se}_{\mathrm{med}}$ the median measured se, L the cell size). Results per site are in Table S7. Driver: `ablation/support_profiles.cjs`.

S9. Third-party toolchains. Six chains rebuild the DTM, contours and simplification from the committed

returns; no chain runs OLV or PDAL code. GDAL 3.13.3: `gdal_rasterize -add` (sum and count), `gdal raster calc`, `fill-nodata`, `proximity`, `gdal_contour`, `gdal vector simplify` (GEOS topology-preserving DP). GRASS 8.5.0: `r.in.xyz` (mean, n), `r.fillnulls` (bilinear), `r.grow.distance`, `r.contour`, `v.generalize method=douglas`. SAGA 7.8.2: Shapes to Grid (mean, count), Close Gaps, Proximity Grid, Contour Lines from Grid, Line Simplification. WhiteboxTools 2.4.0: VectorPointsToRaster (sum, number), FillMissingData, EuclideanDistance, ContoursFromRaster (no smoothing or generalization); lines simplified with scikit-image `approximate_polygon`, since WhiteboxTools has no line simplifier. R 4.6.1: lidR 4.3.3 `pixel_metrics` and `rasterize_terrain` (TIN), terra `distance`, isoband `isolines`, sf `st_simplify`. Python: NumPy binning, SciPy `griddata` (linear) and `distance_transform_edt`, scikit-image `find_contours` and `approximate_polygon`. Contour interval and simplification tolerance equal the cell size.

A script then reads each tool's own outputs. Every source edge is assigned the 2×2 block of cell centres containing its midpoint, and the script checks that the block brackets the contour level (all 23,566 edges did). Provenance is M, I or X from the block's return counts; the measured channel is the smallest return count among measured cells and the interpolation channel the largest distance to data among empty cells. Simplified lines are matched to their source lines by level, end vertices and vertex subsequence, with backtracking where a line passes through the same point twice; every simplified line matched. Because only the order of counts and distances is used, the promotion counts hold for any support function increasing in return count and decreasing in distance.

The tools bin returns by four rules, recorded per chain in the output: half-open cells counted from the south (NumPy; the production code also keeps returns lying exactly on the outer grid boundary, which differs from NumPy in 4 cells at White Sands and Estonia); a return exactly on a horizontal cell edge assigned to the cell above (GDAL, GRASS, WhiteboxTools); the same after rounding coordinates to the 0.001 LAS scale and keeping returns on the outer grid boundary (lidR); and half-open cells restricted to the extent between outer cell centres (SAGA). Re-binning the returns with each rule reproduces that tool's cell counts in every cell at every site. Three WhiteboxTools 2.4.0 defects were worked around: `VectorPointsToRaster` fails with `assign=mean` or `--nodata`, so the mean is sum divided by number; its ASCII export writes the lower-left corner one cell too far north, so GeoTIFF outputs are read and their georeference checked; and the Python wrapper reports success when a tool fails, so every output file is checked. Per-chain results are in Table S8. Driver: `toolchain-audit/reproduce.sh`.

S10. Worked example and attribute query. The query keeps a span when its exported measured support is at least t and its exported provenance is $\{M\}$, for $t = 66$ (solid) and $t = 33$ (dashed). The reference answer applies the same test to each source edge. False acceptance is the length of source edges that fail the test inside spans the query accepts; over-rejection is the length of source edges that pass inside spans it rejects. The traced span is selected by a fixed rule: among DP spans whose endpoint-attributed record passes the query at $t = 66$ and promotes support, the largest promotion, then the longer span, then site order. Seven spans qualified. The traced span is 4.81 m long. Its weak cell holds 2 returns (site median 12), both near its eastern edge, so its support is $C_M(2; 12, 3) = 0.07$; three of the five source segments, 2.62 m in all, depend on that cell. The map draws the stretch as a gap because display grade comes from a segment's weaker end vertex and every segment here touches the weak cell. The support profile of the span reads 45.5% of length at or above 66 and 54.5% below 33. Results per site are in Table S9: at $t = 33$, 4.0–50.8% of the length DP accepted failed at source level, and EC-DP accepted no failing length. Driver: `ablation/worked_example.cjs`.

S11. Directional classifier and closed rings. Directional support is evaluated on eight rays around a cell. A pattern is one-sided when it has a single occupied direction or when the largest angular gap between adjacent occupied rays exceeds four of eight sectors. Because the rule depends on cyclic gaps rather than compass labels, it is invariant under 45-degree rotation; all 256 patterns are verified exhaustively. Closed rings are cut at the lexicographically smallest (x, y) vertex as described in S1. *Chords and unsupported cells.* EC-DP splits a line at unsupported source segments, so no simplified segment replaces one. This does not by itself prevent a chord from passing over an unsupported cell that its source segments went around, and geometry equivalence with DP holds whether or not it does. The case was counted: each one-cell DP chord was tested by exact segment–cell intersection against every cell with no supported elevation; passing through the interior

of such a cell counts, touching only its edge or corner does not. On the two crops that have such cells (White Sands, 32% of cells; Marsh Island, 49%), 0 of 268 chords passed through one; Estonia Tava and StREAM have none to cross. This is an observation on these crops at this tolerance, not a property of the method. A pipeline that needs the stronger guarantee must test each chord against the mask and keep the vertices of a chord that fails. Driver: `ablation/chord_mask.cjs`.

S12. EC-wrap. Given a source line with one evidence record per edge and a simplifier's output whose vertices are a subsequence of the source vertices with both ends kept, EC-wrap recovers the source indices of the output vertices (in order; with backtracking where a line passes through the same point twice) and assigns each output segment the complete-source minimum per channel, the provenance union and the support profile of the edges between consecutive retained indices. The simplifier is a black box. The matching is the one used for the six toolchains (Methods S9). As a non-Douglas–Peucker demonstration, the strict feature runs of the four crops were simplified by Visvalingam–Whyatt (`simplification` 2.0.0, Rust implementation) at an effective area of one source cell squared; every output line matched its source. Smoothing is not a subsequence operator, so EC-wrap does not apply to it; the reference smoothing operator gives each new edge the minimum and provenance union of the parent edges it spans (`smoothEvidence` in `js-study/reference/typedEvidenceOperators.ts`), and EC-DP then acts on the smoothed line. Simplified vertices are matched to source vertices by coordinates rounded to 10^{-9} m. Driver: `ablation/ecwrap.py` with `ablation/export_runs.cjs`.

Site	Lines	Reduction	Spans	Endpoint promotion	Endpoint prov. loss	EC-wrap prom./loss
White Sands	43	65.4%	107	18 (16.8%)	1 (0.9%)	0/0
Estonia Tava	89	62.6%	173	11 (6.4%)	11 (6.4%)	0/0
StREAM	33	63.7%	64	11 (17.2%)	0 (0.0%)	0/0
Marsh Island	72	83.0%	277	11 (4.0%)	0 (0.0%)	0/0

Further simplifiers. Rules fixed before the run (`external-audit/PREREGISTRATION.md`, part A). The same runs were simplified at a tolerance of one source cell by QGIS 3.40.5 (`native:simplifygeometries`; its Douglas–Peucker option calls GEOS 3.9.1, its Visvalingam option is QGIS's own code) and by mapshaper 0.6.102 (its own Douglas–Peucker and Visvalingam, planar mode), and scored as above. EC-wrap's records showed no promotion and no provenance loss in any of the 2,117 spans. mapshaper deletes closed lines that collapse at this tolerance ("removed" below); those lines are not scored. One further mapshaper line was not a vertex subsequence of its source and is left out. Driver: `ablation/ecwrap_tools.py`.

Site	Simplifier	Lines	Removed	Reduction	Spans	Endpoint promotion	Endpoint prov. loss	EC-wrap prom./loss
White Sands	QGIS DP	43	0	65.8%	105	20 (19.0%)	0 (0.0%)	0/0
White Sands	QGIS VW	43	0	64.4%	111	18 (16.2%)	1 (0.9%)	0/0
White Sands	mapshaper DP	34	9	73.2%	69	20 (29.0%)	0 (0.0%)	0/0
White Sands	mapshaper VW	34	9	73.4%	68	22 (32.4%)	1 (1.5%)	0/0
Estonia Tava	QGIS DP	89	0	46.7%	284	8 (2.8%)	9 (3.2%)	0/0
Estonia Tava	QGIS VW	89	0	61.4%	181	11 (6.1%)	11 (6.1%)	0/0
Estonia Tava	mapshaper DP	41	47	66.7%	103	7 (6.8%)	8 (7.8%)	0/0
Estonia Tava	mapshaper VW	37	52	71.6%	75	8 (10.7%)	6 (8.0%)	0/0
StREAM	QGIS DP	33	0	60.7%	72	11 (15.3%)	1 (1.4%)	0/0
StREAM	QGIS VW	33	0	62.5%	67	11 (16.4%)	0 (0.0%)	0/0
StREAM	mapshaper DP	26	7	69.2%	43	11 (25.6%)	1 (2.3%)	0/0
StREAM	mapshaper VW	25	8	71.2%	37	12 (32.4%)	0 (0.0%)	0/0
Marsh Island	QGIS DP	72	0	83.5%	266	8 (3.0%)	0 (0.0%)	0/0
Marsh Island	QGIS VW	72	0	82.7%	284	11 (3.9%)	0 (0.0%)	0/0
Marsh Island	mapshaper DP	45	27	89.4%	158	8 (5.1%)	0 (0.0%)	0/0
Marsh Island	mapshaper VW	45	27	87.5%	194	11 (5.7%)	0 (0.0%)	0/0

S13. Horizontal position uncertainty. Rules fixed before the run. Each cell's vertical standard error se is that of the standard-error profile (Methods S8). For the 2×2 block of a contour edge, the horizontal position uncertainty is $u = \max(\mathrm{se})/\max(|\nabla z|,\ 0.05)$, with $|\nabla z|$ the magnitude of the block's bilinear gradient and 0.05 m/m a floor that keeps flat ground finite. A DP span's true u is the largest over the edges it replaces; its endpoint record carries the larger of its two end edges. The last column repeats the share for spans whose every block lies above the floor, so the floor does not drive the result. Driver: `ablation/positional_uncertainty.cjs`.

Site	Length understated	Median factor	Max factor	Median (m)	Max (m)	Above floor only
White Sands	56.0%	1.26	4.16	0.05	0.36	50.3%
Estonia Tava	32.2%	1.41	2.65	0.41	0.87	24.3%
StREAM	40.5%	1.65	6.37	0.20	1.44	32.6%
Marsh Island	28.8%	1.20	2.46	1.20	5.73	35.2%

S14. Attribute schemas of national contour products. A desk study, read on 9 November 2026, of the published attribute schema of the contour-line class in ten national topographic products (Table S12). Only attributes stored on each contour feature were counted; dataset-level metadata were excluded. Each schema was read from the agency's own specification, feature catalogue or service definition, at the address listed. Four questions were asked of each schema: does a contour carry (a) the kind of source or the way it was generated, (b) a quality flag, (c) an accuracy value, (d) a link to its source dataset. No schema stores any of these below the level of the whole feature, and none stores a support value: each quality flag is a category or yes/no, and each accuracy value is one number for the whole line. Five schemas record a source or generation type, three a quality flag, five an accuracy value and six a source link, in full or in part. Four schemas (United States, Great Britain, Sweden, Japan) carry none of (a)–(c). The national products of France and Estonia were also read; neither publishes a vector contour class, so they are not in the table. The survey describes published schemas, not the content of delivered files, and ten products are not a census.

S15. Held-out ground check. Rules fixed before the run and kept in the driver's header. Each crop's ground returns were split at random into two halves, twenty times (seeds 1–20). Terrain, support, contours and one-cell DP were built from one half; the other half was held out. The terrain therefore has half the paper's return density, and its counts are not mixed with the main tables. For a cell holding held-out returns the error is $e = |\bar{z}_{\text{held}} - z_{\text{terrain}}|$; the error of a contour edge is the RMS of e over the cells of its 2×2 block that hold held-out returns. Three questions were asked. (1) Does support order error? RMSE of e by cell class (measured cells with support ≥ 66, 33–66 and < 33; interpolated cells) and Spearman's ρ between support and e. (2) On DP spans whose endpoint record is promoted, is the error larger on the replaced edges the record hides than on the two end edges it reads? (3) Among spans the endpoint record grades solid, is the largest edge error greater on those EC-DP downgrades than on those it keeps? Every value is the median over the 20 splits, with the range in brackets.

Site	Cell class	Cells	RMSE of e (m)
White Sands	Measured, ≥ 66	2159	0.021 [0.020–0.022]
	Measured, 33–66	899	0.022 [0.020–0.024]
	Measured, < 33	432.5	0.026 [0.025–0.030]
	Interpolated	17.5	0.061 [0.039–0.097]
Estonia Tava	Measured, ≥ 66	0	n/a
	Measured, 33–66	380.5	0.047 [0.042–0.052]
	Measured, < 33	1791	0.051 [0.050–0.052]
	Interpolated	2715.5	0.048 [0.046–0.050]
StREAM	Measured, ≥ 66	5	0.026 [0.009–0.097]
	Measured, 33–66	219	0.060 [0.051–0.071]
	Measured, < 33	254	0.087 [0.070–0.098]
	Interpolated	297.5	0.174 [0.151–0.191]
Marsh Island	Measured, ≥ 66	896	0.005 [0.004–0.006]
	Measured, 33–66	231	0.009 [0.007–0.018]
	Measured, < 33	386	0.013 [0.012–0.015]
	Interpolated	32	0.018 [0.011–0.226]

		Promoted spans with held-out data on hidden and end edges		
Site	Spearman ρ (support, e)	Spans	Hidden error larger	Hidden/end error, median ratio
White Sands	−0.04 [−0.08 to −0.02]	25.5	58% [44%–70%]	1.07 [0.94–1.31]
Estonia Tava	0.01 [−0.01 to 0.02]	9.5	50% [22%–91%]	0.99 [0.86–1.40]
StREAM	−0.05 [−0.10 to 0.01]	6	50% [17%–100%]	1.03 [0.52–2.03]
Marsh Island	−0.42 [−0.47 to −0.40]	6.5	63% [33%–88%]	1.23 [0.85–2.31]

(1) Median RMSE rose from the highest to the lowest support class of measured cells at every site, and interpolated cells had the largest error at three sites; at Estonia Tava, which has no cell at or above 66, they fell between the two measured classes. Cell by cell the relation is weak: ρ was clearly negative only at Marsh

Island and within 0.1 of zero elsewhere. (2) Inconclusive. A split left 6–26 promoted spans per site with data on both kinds of edge; the hidden edges had the larger error in 50–63% of them, and every range of the ratio includes 1. (3) Inconclusive. Only White Sands had spans in both groups on every split (9 kept, 5 downgraded): median error was 0.029 m on the downgraded spans against 0.023 m on the kept ones, a ratio of 1.33 [0.78–2.31]. Marsh Island had a downgraded span on two splits; Estonia Tava and StREAM had no span graded solid. The check supports the ordering of support by cell class. At span level it neither confirms nor refutes a difference in ground error: 100 m crops at half density hold too few promoted spans. Driver: `ablation/heldout_check.cjs`.

The same check on the scale-test crops. Rules fixed before the run (`docs/RULES_ADDITIONAL_TESTS.md`, D2). The four crops gave too few promoted spans to answer question 2, so the same definitions were applied to the 129 scored crops of Methods S18, with seeds 1–5. For each promoted span with held-out data on a hidden edge and on an end edge, d is the mean hidden-edge error minus the mean end-edge error; spans are pooled over crops and S is the share with $d > 0$. The interval is from 2,000 bootstrap replicates that resample whole crops. Reading fixed beforehand: an interval wholly above 50% means the hidden segments have larger held-out error. S was 57.9–58.6% over the five seeds, each interval above 50%, so the segments an endpoint record hides had larger held-out error than the segments it reads. The difference is small: a median of about 3 mm, a ratio of 1.09. RMSE rose from one measured class to the next in about three quarters of the eligible crops, and ρ was negative but weak. Among spans the endpoint record grades solid, those EC-DP downgrades had the larger error (question 3). Values are medians over seeds. Driver: `scale/heldout_scale.cjs`.

Crops	n	Spans	S [95%]	Median d (mm)	Ratio	RMSE rises	ρ median [quartiles]	Q3 share [95%]
All scored	129	7,048	58.3% [56.9–59.6]	2.9	1.09	90/119	−0.13 [−0.19 to −0.07]	74.0% [62.5–88.5]
Without parent tiles	113	6,377	58.7% [57.2–60.1]	3.1	1.09	86/112	−0.14 [−0.20 to −0.09]	79.0% [59.6–92.5]

S16. Fewest vertices for a geometry-only fix. Rules fixed before the run. A geometry-only fix keeps endpoint records and changes only which vertices are retained. On the feature lines used for every baseline (Methods S7), a span between two source vertices is allowed if every source vertex between them lies within one cell of the line through its ends, the distance rule of the production DP. The fewest vertices that can be retained under a condition on spans is the shortest path from a line's first vertex to its last over allowed spans that meet the condition, found exactly by dynamic programming. Three conditions were used: none (the fewest vertices any selection within tolerance can keep); no promotion in any channel and no provenance class lost by the endpoint record; and exact records, where every replaced edge has the provenance and support of the span's record, which is what homogeneous splitting enforces by cutting.

				Fewest vertices within tolerance			
Site	Lines	Source	DP	No condition	No promotion or loss	Exact records	Homogeneous splitting
White Sands	43	433	125	118 (0.94)	129 (1.03)	288 (2.30)	288 (2.30)
Estonia Tava	89	700	261	250 (0.96)	268 (1.03)	462 (1.77)	462 (1.77)
StREAM	33	267	87	85 (0.98)	95 (1.09)	207 (2.38)	207 (2.38)
Marsh Island	72	2052	258	242 (0.94)	248 (0.96)	605 (2.34)	609 (2.36)

Retained vertices, with the multiple of DP in parentheses. DP kept 2–7% more vertices than the fewest possible. Homogeneous splitting equalled the exact-record minimum at three sites and kept four more vertices at Marsh Island, so its cost (1.77–2.38 times DP) belongs to the requirement of exact endpoint records, not to the splitting procedure. Avoiding only promotion and provenance loss needs 0.96–1.09 times the DP vertices. Such a selection is a different simplification: its lines differ from DP's wherever DP's endpoint records were promoted, it must read the evidence while it simplifies, and its endpoint records then equal the complete-source records for that geometry. EC-DP reaches complete-source records on the unchanged DP geometry. Driver: `ablation/optimal_bound.cjs`.

S17. Sensitivity of the position-uncertainty result. Methods S13 fixes two constants, an instrument precision of 0.05 m and a slope floor of 0.05 m/m. Rule fixed before the run: repeat S13 with each constant at 0.02, 0.05 and 0.10, nine combinations per site.

Site	Length understated, S13 values	Range over nine combinations	Median factor, range
White Sands	56.0%	43.8–60.0%	1.23–1.43
Estonia Tava	32.2%	19.1–46.1%	1.11–1.95
StREAM	40.5%	37.3–40.8%	1.44–1.89
Marsh Island	28.8%	12.2–38.0%	1.08–1.23

Endpoint records understated the worst horizontal uncertainty on 12.2–60.0% of contour length over all sites and combinations; no combination removed the effect. Driver: `ablation/positional_sensitivity.cjs`.

S18. Scale test on public-agency tiles. Rules written on 9 November 2026 before any crop was cut (`scale/README.md`). *Tiles*: every airborne LiDAR tile from a public-agency programme on the author's disk on that date, 18 in all: 8 USGS 3DEP tiles (Arizona, Coconino 2019), 4 USGS tiles (Wisconsin, Racine and Kenosha 2017), 4 NOAA/USGS tiles (Hawaii 2020–2023), and the USGS White Sands and Estonian Land Board tiles from which two of the paper's crops were cut. They had been downloaded for other work and are a convenience sample, not a random one. *Crops*: nine per tile, 100 m by 100 m, centred at 1/4, 1/2 and 3/4 of the tile's extent in each axis; ground returns only. *Processing*: as for the White Sands and Estonia crops (1 m grid, 1 m contour interval, one-cell DP), on the feature lines of Methods S7. *Exclusions*: a crop with fewer than 1,000 ground returns is skipped and counted; per-crop rates are given for crops with at least 20 DP spans, and pooled counts include every scored crop.

One amendment was written after the crops were cut and before any was scored. Two Hawaii tiles are small, so some of their nine crops overlap; within a tile, a crop that overlaps one already kept is dropped and counted. Cutting also showed that both Kenosha tiles hold ground returns in one corner only, which no crop reaches, and that two White Sands crops lie outside the data; these fall under the 1,000-return rule, and no crop was moved to find data. Quartiles use linear interpolation between order statistics.

		Crops		Pooled DP spans, endpoint records			Promotion per crop	Splitting
Survey	Tiles	Scored	Dropped	Spans	Promoted	Provenance loss	median [quartiles] (range), %	vertices
Arizona, Coconino	8	72	0	15,702	6,114 (38.9%)	1,398 (8.9%)	37.2 [31.7–42.7] (14.6–60.9)	×2.56
Wisconsin	4	18	18	2,106	534 (25.4%)	15 (0.7%)	24.6 [18.9–41.6] (10.7–50.0)	×2.07
Hawaii	4	23	13	10,489	2,467 (23.5%)	704 (6.7%)	23.2 [21.3–25.2] (14.3–27.9)	×2.32
White Sands tile	1	7	2	1,686	406 (24.1%)	2 (0.1%)	26.7 [19.2–26.8] (18.1–28.7)	×1.93
Estonia Tava tile	1	9	0	2,259	346 (15.3%)	150 (6.6%)	17.1 [12.8–20.6] (5.9–35.0)	×1.94
All	18	129	33	32,242	9,867 (30.6%)	2,269 (7.0%)	30.0 [22.9–40.4] (5.9–60.9)	×2.39
Without paper tiles	16	113	31	28,297	9,115 (32.2%)	2,117 (7.5%)	31.9 [24.1–41.6] (10.7–60.9)	×2.44

Of 162 crops, 13 were dropped for overlap and 20 for too few returns; 129 were scored, 123 of them with at least 20 spans. Endpoint attribution promoted support in 125 of the 129 crops and lost provenance in 101. Per-crop provenance loss had a median of 5.9% (quartiles 1.6–9.0%, range 0.0–18.2%). EC-DP showed no promotion and no provenance loss in any crop. Homogeneous splitting showed no promotion and kept 2.39 times the DP vertices (per crop: median 2.31, range 1.58–3.47). All Wisconsin crops that were scored come from the two Racine tiles. The rates describe these tiles; they are not an estimate of prevalence in any national product. The crops are not in the repository because of their size; the SHA-256 digest of each is in the result file. Drivers: `scale/extract_crops.py`, `scale/scale_test.cjs`.

S19. Contours and points published by other authors. Rules written on 9 November 2026 before any evidence was computed (`external-audit/PREREGISTRATION.md`, part B). *Data*: the contour file and the LiDAR point file of You and Lee (2024), Zenodo, doi:10.5281/zenodo.12771247, CC0, used as published: 1,015 contour lines with 139,271 vertices at a 20 m interval and 96,502 points over about 1.4 km by 1.5 km of mountain terrain. The record does not say how the contours were drawn, and no step of this test uses the author's rasterizer or contouring. *Lines*: 20 lines with fewer than three vertices are left out, leaving 995. *Evidence*: the points are binned on a square grid whose cell is the nominal point spacing (4.47 m) rounded up, 5 m. A cell with points is measured and its support is its point count; an empty cell is interpolated and its support falls with the distance to the nearest measured cell. Support is compared by order only, as in Methods S9. Each published contour segment takes the evidence of the cell under its midpoint (55.7% of segments are measured). *Simplifiers*: Douglas–Peucker and Visvalingam–Whyatt of the Rust library, QGIS and mapshaper (Methods S12), at a tolerance of one cell. *Scoring*: as in Methods S12. *Sensitivity*: cells and tolerance of 2.5 m and 10 m with the Rust Douglas–Peucker.

Endpoint attribution promoted support on 28.0–40.7% of spans and lost provenance on 24.6–35.7% under the six simplifiers at 5 m, and on 39.7% and 43.8% of spans at 2.5 m and 10 m. EC-wrap's records showed neither in any span. mapshaper removed 359 and 363 closed lines that collapsed. The published contour file has four columns (X, Y, Z, line number) and no evidence field. The evidence rule is the author's, applied to data the author did not produce: the result shows that the mechanism operates on external contour geometry and point distributions, and it is not a measurement of how often published products contain it. Driver: `external-audit/external_audit.py`.

Cell (m)	Simplifier	Lines	Removed	Reduction	Spans	Endpoint promotion	Endpoint prov. loss	EC-wrap prom./loss
5	Rust DP	995	0	96.1%	4,474	1,732 (38.7%)	1,533 (34.3%)	0/0
5	Rust VW	995	0	93.5%	8,041	2,260 (28.1%)	1,982 (24.6%)	0/0
5	QGIS DP	995	0	95.4%	5,446	1,695 (31.1%)	1,488 (27.3%)	0/0
5	QGIS VW	995	0	93.5%	8,048	2,250 (28.0%)	1,977 (24.6%)	0/0
5	mapshaper DP	636	359	96.5%	4,179	1,699 (40.7%)	1,490 (35.7%)	0/0
5	mapshaper VW	632	363	95.8%	5,144	1,927 (37.5%)	1,655 (32.2%)	0/0
2.5	Rust DP	995	0	93.6%	7,984	3,166 (39.7%)	2,600 (32.6%)	0/0
10	Rust DP	995	0	97.4%	2,672	1,170 (43.8%)	593 (22.2%)	0/0

S20. Attribute handling when segments are dissolved. Rules fixed before the run (`docs/RULES_ADDITIONAL_TESTS.md`, D1). Each source edge of the production contour runs of the four crops was written as a two-vertex line feature with its evidence (provenance, measured support, interpolation support) and the index of its line. Four tools then joined the features of each line into one feature. QGIS and GeoPandas were run with default settings. mapshaper drops other fields when it dissolves unless `copy-fields` names them, so that option was used. The GDAL/OGR query names the evidence fields as bare columns in a grouped query, for which SQLite does not guarantee which row supplies the value. The calls: QGIS 3.40.5 `native:dissolve`, GeoPandas 1.0.1 `dissolve`, mapshaper 0.6.102 `-dissolve` with `copy-fields`, and GDAL/OGR 3.13.3 with an SQLite `GROUP BY`. The tool writes the record; the script only reads it back. A dissolved line shows support promotion when the tool's value exceeds the minimum over the line's edges in either channel, and provenance loss when a provenance type of its edges is absent from the tool's value. Lines with one edge (23 of 237) are not scored. In this run QGIS, mapshaper and GDAL/OGR wrote the record of the first feature of each line on all 237 lines (for GDAL/OGR an observation, not a guarantee of SQLite); GeoPandas wrote, for each field, the first value that is not null. With features in source order the four workflows promoted support on 26.2–75.0% of lines and lost provenance on 5.6–25.0%; taking the mean of the support fields (GeoPandas `aggfunc="mean"`; OGR `AVG`) promoted on 56.2–92.9%. With features in reverse order the same four calls promoted on 22.5–66.7% and lost provenance on 5.6–46.4%. No tool dropped the fields. Driver: `ablation/dissolve_tools.py`.

Order	Site	Lines	QGIS	GeoPandas	mapshaper	GDAL/OGR	GeoPandas mean	OGR mean
Source	White Sands	36	22 (61.1%) / 2 (5.6%)	22 (61.1%) / 2 (5.6%)	22 (61.1%) / 2 (5.6%)	22 (61.1%) / 2 (5.6%)	33 (91.7%)	33 (91.7%)
Source	Estonia Tava	80	21 (26.2%) / 20 (25.0%)	25 (31.2%) / 20 (25.0%)	21 (26.2%) / 20 (25.0%)	21 (26.2%) / 20 (25.0%)	45 (56.2%)	45 (56.2%)
Source	StREAM	28	19 (67.9%) / 6 (21.4%)	21 (75.0%) / 6 (21.4%)	19 (67.9%) / 6 (21.4%)	19 (67.9%) / 6 (21.4%)	26 (92.9%)	26 (92.9%)
Source	Marsh Island	70	24 (34.3%) / 9 (12.9%)	25 (35.7%) / 9 (12.9%)	24 (34.3%) / 9 (12.9%)	24 (34.3%) / 9 (12.9%)	54 (77.1%)	54 (77.1%)
Reversed	White Sands	36	24 (66.7%) / 2 (5.6%)	24 (66.7%) / 2 (5.6%)	24 (66.7%) / 2 (5.6%)	24 (66.7%) / 2 (5.6%)	33 (91.7%)	33 (91.7%)
Reversed	Estonia Tava	80	18 (22.5%) / 30 (37.5%)	24 (30.0%) / 30 (37.5%)	18 (22.5%) / 30 (37.5%)	18 (22.5%) / 30 (37.5%)	45 (56.2%)	45 (56.2%)
Reversed	StREAM	28	14 (50.0%) / 13 (46.4%)	17 (60.7%) / 13 (46.4%)	14 (50.0%) / 13 (46.4%)	14 (50.0%) / 13 (46.4%)	26 (92.9%)	26 (92.9%)
Reversed	Marsh Island	70	19 (27.1%) / 10 (14.3%)	21 (30.0%) / 10 (14.3%)	19 (27.1%) / 10 (14.3%)	19 (27.1%) / 10 (14.3%)	54 (77.1%)	54 (77.1%)

Cells: support promotion / provenance loss, count (% of lines). The mean columns carry no provenance field.

S21. Raster resampling. Rules fixed before the run (`docs/RULES_ADDITIONAL_TESTS.md`, D3). The production DTM of each crop was written as three aligned rasters: measured support, interpolation support, and a provenance code (measured, interpolated, unsupported as no-data). GDAL 3.13.3 `gdalwarp` resampled them to cells two and four times larger, aligned on the grid origin so that every coarse cell is an exact block of source cells, with nearest-neighbour, average and mode resampling, and with the minimum as the contract's rule for the support rasters. A resampled raster is treated as a representation of the same terrain estimate at coarser cells, so resampling falls under the contract. A coarse cell shows support promotion when a resampled support value exceeds the minimum of that raster over its block, provenance loss when the resampled code omits a provenance type present in the block, and a value over unsupported cells when it carries a value although its block contains an unsupported cell. Blocks with no supported cell are not scored. Nearest-neighbour promoted support in 25.9–71.8% of coarse cells and averaging in 50.9–100.0%; the minimum promoted in none. The provenance code lost a type in 2.3–95.0% of cells under nearest-neighbour and 1.5–95.0% under mode, because a single code cannot hold a mixed block. Every resampling, the minimum included, wrote values over blocks that contain unsupported cells at the two sites that have them, so rule (ii) needs a no-data rule that the resampler does not apply by default. Estonia Tava and StREAM have no unsupported cells. Drivers: `ablation/export_dtm_grids.cjs`, `ablation/resample_gdal.py`.

			Support promotion			Provenance loss		Value over
Site	Factor	Cells	Nearest	Average	Minimum	Nearest	Mode	unsupported (average)
White Sands	2	1,721	954 (55.4%)	1261 (73.3%)	0 (0.0%)	40 (2.3%)	40 (2.3%)	41 (2.4%)
White Sands	4	436	313 (71.8%)	359 (82.3%)	0 (0.0%)	36 (8.3%)	25 (5.7%)	21 (4.8%)
Estonia Tava	2	2,500	855 (34.2%)	2053 (82.1%)	0 (0.0%)	1383 (55.3%)	1383 (55.3%)	0 (0.0%)
Estonia Tava	4	625	321 (51.4%)	625 (100.0%)	0 (0.0%)	584 (93.4%)	584 (93.4%)	0 (0.0%)
StREAM	2	400	179 (44.8%)	370 (92.5%)	0 (0.0%)	223 (55.8%)	223 (55.8%)	0 (0.0%)
StREAM	4	100	71 (71.0%)	100 (100.0%)	0 (0.0%)	95 (95.0%)	95 (95.0%)	0 (0.0%)
Marsh Island	2	25,646	6636 (25.9%)	13055 (50.9%)	0 (0.0%)	770 (3.0%)	388 (1.5%)	653 (2.5%)
Marsh Island	4	6,612	3768 (57.0%)	5092 (77.0%)	0 (0.0%)	601 (9.1%)	306 (4.6%)	541 (8.2%)

S22. Probability sample of USGS 3DEP work units. Rules written before any sample crop was fetched (`docs/RULES_ADDITIONAL_TESTS.md`, D4). *Frame*: the 2,283 work units of the public index of USGS 3D Elevation Program point clouds (`boundaries/resources.geojson` of `github.com/hobuinc/usgs-lidar`, commit `ba8a11d1`; U.S. Government public domain). *Draw*: the units were sorted by name and shuffled once with a seeded generator, and taken in that order until 60 valid crops were obtained. *Crop*: one point drawn uniformly in longitude and latitude inside the unit's boundary; the 100 m square centred on it in the UTM zone of its longitude; class-2 ground returns read from the unit's public Entwine Point Tile resource and reprojected; up to three points per unit. *Valid crop*: at least 1,000 ground returns and a determined vertical unit. The point-tile resources do not state the vertical unit, so the crop's median elevation was compared with the USGS Elevation Point Query Service at the crop centre, read as metres and as feet; the nearer reading was taken when it was nearer by at least 5 m, and the unit was otherwise set aside. *Scoring*: as Methods S18.

62 units were tried for 60 valid crops (58 in metres, 2 in feet, converted). Not used: USGS_LPC_CA_SanDiegoQL1_2014_LAS_2016 (vertical unit undetermined); CA_Scripps-Mar_2003 (fewer than 1,000 ground returns). Crop medians agreed with the reference elevation to a median of 0.27 m. Endpoint records promoted support in 59 of 60 crops (98.3%; Wilson 95% interval; with 60 of 2,283 units the finite-population correction is 0.99 and is not applied; 91.1–99.7%). Among the 56 crops with at least 20 spans the promotion rate had a

median of 20.6% (quartiles 13.5–33.1%, range 1.6–55.9%) and a mean of 23.0% (95% interval from 2,000 bootstrap replicates over crops, 19.8–26.3%); pooled, 3,928 of 14,018 spans (28.0%). EC-DP promoted none of the 14,058 spans of all valid crops. The one crop without promotion (Barataria, Louisiana) had two spans. The sample speaks for work units with classified ground returns and a determinable vertical unit, with one 100 m crop per unit; the evidence model and the scoring are this study's, and the result is the frequency of the conditions under which endpoint attribution promotes, not of errors in published products. Drivers: `prevalence/draw_sample.py`, `prevalence/score_sample.cjs`.

Work unit	Returns	Spans	Promoted	Work unit	Returns	Spans	Promoted
TX_Victoria_2006	5,808	16	2	FL_Peninsular_FDEM_Okeechobee_2018	83,998	13	2
USGS_LPC_MN_Phase1_LeSueurCO_2010_LAS_2016	5,223	268	31	WY_Southwest_4_2020	37,742	253	51
USGS_LPC_WY_Casper_Natrona_2015_LAS_2017	36,801	103	19	AR_UpperWhiteVillage-FEMA_2012	4,078	187	34
KS_Osage_2010	4,874	92	27	USGS_LPC_TX_South_B2_2018_LAS_2019	28,947	74	12
WI_Oshkosh_3Rivers_FondDuLac_TL_2018	36,051	243	58	ME_SouthCoastal_2_2020	60,293	492	181
KS_MiamiCo_2009	4,763	119	23	AK_EMRI_2_D24	27,010	363	87
WV_FEMAR3_Southcentral_B1_2018	77,036	974	451	NM_SouthCentral_Acoma_TL_2018	17,834	512	158
MO_LivingstonCounty_2011	9,061	505	19	USGS_LPC_OK_Woodward_UTM15_B5_2016_LAS_2018	53,443	252	80
AL_SWCentral_1_B22	14,991	183	21	KY_FluorsparDistrict_2019	44,139	75	25
USGS_LPC_IL_District4_Tazewell_2014_LAS_2016	9,588	108	18	CA_SanJoaquin_6_2021	144,663	9	3
ID_DryCreek_2007	14,129	705	233	USGS_LPC_MI_31Co_Sanilac_2017_LAS_2019	16,019	70	2
USGS_LPC_MI_31Co_Jackson_2016_LAS_2019	22,954	167	29	MI_FEMA_Cheboygan_2019	23,868	225	25
AL_19Co_1_B24	70,493	269	95	WY_YellowstoneNP_1_2020	20,944	145	20
TX_MillsCo_2006	3,731	132	16	USGS_LPC_MO_FEMAR7_North_A2_2017_LAS_2019	46,219	653	293
SC_Berkeley_2016	32,436	90	13	USGS_LPC_TN_3County_2015_LAS_2017	28,315	278	76
USGS_LPC_TX_West_Central_B7_2018_LAS_2019	33,590	59	6	WV_FEMAHQ_B2_2018	31,009	567	173
USGS_LPC_IN_Central_MarionCo_2016_LAS_2018	12,673	149	21	USGS_LPC_TX_West_Central_B11_2018_LAS_2019	49,942	494	183
USGS_LPC_IN_ET_B6_Jay_2012_LAS_2016	5,921	101	11	USGS_LPC_IN_WT_B10_Benton_2013_LAS_2016	4,972	205	37
FL_LeonCo_2009	10,522	169	32	USGS_LPC_MI_31Co_Midland_2016_LAS_2019	18,248	125	3
USGS_LPC_AZ_BrawleyRillito_NRCS_2018	53,665	179	33	GA_Statewide_B2_2018	8,312	166	59
USGS_LPC_MI_Wexford_2016_LAS_2017	19,058	164	37	ND_3DEPProcessing_4_D22	21,158	128	2
NE_Southwest_1_2020	52,918	133	45	USGS_LPC_MO_AR_Miss_Laud_UTM16_2014_LAS_2016	18,269	134	12
WA_EasternCascades_3_2019	135,377	145	81	TN_OSMRE-GPSC_2011	10,648	235	99
ID_CityofRocks_2011	22,114	283	95	NM_HermitsPeak_1_D23	143,699	781	369
USGS_LPC_OK_DamRehab3_7_2011_LAS_2018	4,267	49	5	USGS_LPC_MO_AR_Mercer_Grundy_2014_LAS_2016	12,396	143	30
SD_MO_River_Dewberry_B2_2016	109,649	48	6	ME_CrownofMaine_B2_2018	21,805	468	126
CA_OrangeCo_2011	14,633	264	43	USGS_LPC_LA_Barataria_2013_LAS_2015	2,206	2	0
MO_StLouis_2012	8,105	212	8	CT_Statewide_B5_2016	23,208	261	58
NE_SouthernNE_B4B_2018	57,460	189	78	GA_Central_4_2018	20,643	224	57
ARRA-FL_Charlotte-W_2010	11,975	264	69	OK_Panhandle_B1C_2018	33,250	142	53

Table S3. Notation summary for the EMTRF contract (main-text Section 2).

Symbol	Meaning
$e_i = (p_i, S_i, A_i, \ell_i)$	evidence record for cell i
$p_i \subseteq \{M, I\}$	provenance: measured-derived M, interpolated-derived I; mixed $X = \{M, I\}$
U	unsupported state: no elevation authorized; no output may replace or join across it, and no later step can make it supported
S_i	support values, one per channel, in $[0, 1]$
A_i	applicability and scientific scope of each support channel
ℓ_i	source lineage
μ	a support channel (e.g. `measured-support`)
D_y; D_y^{μ}	complete required source set for output y; its subset for which μ is applicable
$S_y(\mu) = \min_{x \in D_y^{\mu}} S_x(\mu)$	minimum within one channel
$C_M(n; n_t, h)$; $C_I(d, \rho, \gamma)$	reference measured / interpolated support profiles
p_j^{EC}; $S_j^{\mathrm{EC}}(\mu)$	EC-DP provenance union and support minimum over all replaced segments
n; m; ε	source vertices; retained segments; Douglas–Peucker tolerance

Supplementary Notes

N1. Worked substitution of the reference support profile. The measured and interpolated reference support functions (main-text Eqs. 5–6) are conformance profiles, not part of the EMTRF contract, and may be replaced by any monotone support model without affecting provenance, applicability, lineage, or EC-DP geometry. As a concrete example, suppose an interpolation method supplies a kriging standard error σ_K and defines a new

channel $S_K = \sigma_0/(\sigma_0 + \sigma_K)$ with $\sigma_0 > 0$, applicable only to interpolated-derived cells. For $\sigma_0 = 0.25$ m and a source arc with $\sigma_K = \{0.10, 0.25, 0.40\}$ m, the channel values are approximately $\{0.714, 0.500, 0.385\}$ and the minimum over the arc is 0.385. Replacing the cell-level formula changes only how that channel is built; the provenance union, the rule that no output replaces an unsupported source segment, applicability, lineage, and EC-DP geometry are unchanged. This substitutability is why the reported mechanism failures do not depend on the particular numerical support scale.

N2. Proofs. *Proposition 1.* For a fixed channel μ with every applicable source supplying a value, $S_y(\mu) = \min_{x \in D_y^\mu} S_x(\mu)$ is bounded above by each $S_x(\mu)$ by definition of the minimum, and commutativity, associativity and idempotence are those of min on $[0, 1]$. Unlike channels are never combined, and a missing applicable value is not skipped: by rule (iv) the channel is unavailable.

Proposition 2. EC-DP runs the same recursion and distance test as ordinary DP and attaches evidence only after the retained index set is fixed, so on a fully supported line at fixed tolerance, with no caller-supplied protected indices, the retained sets coincide. The equality is also asserted executably, including across provenance transitions.

Proposition 3. For segment j and applicable channel μ with complete values, EC-DP assigns $\min_{k \in D_j^\mu} S_k(\mu)$, which is at most $S_k(\mu)$ for every applicable k on the replaced arc; with an applicable value missing it emits no number.

Theorem 1. Consider operators $T_1, \ldots, T_r$, each obeying rules (i)–(vi), and an output z of the chain with ancestor set D_z^* (sources of sources, by composition of the per-operator source maps). (i) Union is associative, so the provenance of z is the exact union over D_z^*; nothing is removed or invented. (ii) If any required ancestor is unsupported, the operator receiving it emits no supported element across it, so no later operator receives one. (iii) For channel μ, each operator emits either no value or a value bounded by every applicable input value; by transitivity the final value is bounded by every applicable ancestor value. Sources for which μ is not applicable are excluded at the step where they enter and so never bound or contribute. A channel applies to an output exactly when it applies to at least one of its required sources; if it applies to none, the output carries no value and no applicability for μ, so at later steps it neither contributes a value nor blocks one, and any later value is bounded by the applicable ancestors that do carry μ. (iv) An unavailable channel at step s is, for step $s + 1$, an applicable source with a missing value, so rule (iv) at step $s + 1$ keeps it unavailable; by induction it stays unavailable to the end of the chain and is never replaced by a value computed from other sources. (v) Scope is an intersection at each step, and repeated intersection cannot broaden it. (vi) Lineage at each step identifies the complete source set, so composing the maps identifies D_z^*.

N3. Synthetic-surface GDAL audit. Contours from GDAL `gdal_contour` 3.13.1 were extracted at 0.5 m from a deterministic analytic DEM and scored against a reference support surface that decays with distance from two measurement passes. The output schema carried elevation and identifier fields but no provenance, support, applicability or lineage field. Endpoint-only inheritance promoted 18 of 91 spans (19.8%); complete-source inheritance promoted none. The audit runs in the reproduction gate (`gdal-audit/`); the main text reports the six-toolchain audit on the four field crops instead (Methods S9).

N4. Support-profile record. For a span representing source edges with lengths l_k and channel values v_k, the profile stores $f_t = \sum_{k: v_k \geq t} l_k / \sum_k l_k$ for each display threshold $t \in \{66, 33\}$, beside $\min_k v_k$. Merging spans j with represented source lengths L_j (recoverable from the lineage interval) gives $f_t = \sum_j f_{t,j} L_j / \sum_j L_j$ exactly when the merged spans partition the represented source length, as consecutive EC-DP spans do; spans that share source edges need those edges counted once. Under that condition the record is exact under composition and cannot overstate at any step. The profile gives the total source length at or above a threshold in one channel. It does not say where along the chord that length lies, and it does not answer a query that also tests provenance: at White Sands and $t = 33$ the measured channel has 241.52 m at or above the threshold, of which 237.46 m is on measured-only edges (Tables S9, S11). Scalar bound: a single record r safe for every threshold query must satisfy $r \leq \min_k v_k$, otherwise the query at any $t \in (\min_k v_k, r]$ accepts the span while the minimum edge fails it. The driver `ablation/evidence_profile.cjs` checks all three statements on the four crops: profile length accounting exact in every site and channel (Table S11); two-cell spans equal to the length-weighted merge of their one-cell spans in 708 of 708 span–channel checks; and an unsafe threshold found for all 67 DP

spans whose endpoint record exceeds the arc minimum. The record needs three numeric attribute fields per channel.

Table S4. Source context for the four field-derived contour crops. Metadata are limited to the fields frozen in the current OLV terrain-field and dataset manifests; "not frozen" means the committed crop records campaign/year context but not an exact flight day or instrument model. Checkpoints, where present, are not used to validate the EMTRF representation invariant.

Crop	Acquisition / platform	Instrument and committed crop context	CRS / vertical reference
White Sands	2020 USGS 3DEP airborne LiDAR	Optech Galaxy; gypsum-dune ground crop derived from source tile `USGS_LPC_NM_WhiteSandsNM_2020_D20_w3597n3635.laz`	EPSG:6342 / NAVD88 Geoid12B
StREAM	23 April 2026; two UAV LiDAR flights during leaf-off conditions	DJI Matrice 350 RTK + Zenmuse L1; riparian field, road, trees, and stream corridor; 0.1 m DTM/CHM source products published with the dataset	EPSG:6346 / NAVD88
Estonia Tava	2020 national airborne LiDAR; exact flight day not frozen in crop metadata	National ALS source; 12 571-point class-2 ground crop, flat boreal plain with about 1.6 m relief in the committed crop	EPSG:3301 / EH2000
Marsh Island	March 2025 campaign; site 2025-009-FA; exact flight day is not frozen in the committed crop manifest	UAS LiDAR product; flat coastal salt marsh; Emlid RS3 RTK checkpoints exist in the release but are outside the EMTRF invariant test	EPSG:6348 / NAVD88 GEOID18

Table S5. Claim–evidence boundary used throughout the study. The rows separate representation conformance, empirical mechanism occurrence, upstream numerical agreement, and terrain accuracy so evidence from one row is not used to authorize another.

Claim class	Evidence used here	Not established by that evidence
Operator conformance	Propositions; exhaustive directional states; adversarial source arcs/gaps; JavaScript–Python differential check	DTM vertical accuracy; population defect prevalence
Mechanism on field geometry	Four public contour crops; complete-source audit; polyline-cluster bootstrap sensitivity summaries; P0–P3 ablation	Balanced terrain/sensor effects; population rates
Upstream numerical implementation	Independent SciPy DTM and GDAL contour E4 records in the audited source	Correct export provenance; field ground truth; EMTRF conformance
Application counterexample	Frozen source-path audit of grade-derived export evidence	Frequency of the same defect in other GIS packages
Third-party geometry	Six open-source toolchains (GDAL, GRASS, SAGA, WhiteboxTools, lidR, SciPy/scikit-image) building DTM, contours and simplification on the four crops (Methods S9)	Population prevalence across GIS tools; correctness of the author-written attribution step beyond its recorded checks
Alternative methods and support functions	Boundary-vertex and homogeneous-splitting baselines; exponential, kriging and standard-error support profiles (Methods S7–S8)	Optimality of the minimum rule; calibrated probabilities
Runtime	Paired synthetic and real-contour representation-stage benchmarks	Universal browser/native-GIS or end-to-end pipeline overhead

Table S6. Orthogonal cross-implementation records bundled with the audited v0.6.5 source. E4 denotes independent implementation agreement in the application's evidence ladder, not external field truth.

Product	Independent reference	Executed scope	Compared	Max. difference
DTM mean rasterization	SciPy binned point-in-cell mean (reference stored to 0.1 mm)	Estonia Tava, 1 m grid; identical data/NODATA mask	7,627 cells	3.6×10^{-5} m
Contour placement	GDAL 3.13.1 `gdal_contour`	Tilted-plane DEM; 0.5 m interval	2,222 vertices	2.90×10^{-5} m

Table S7. Support-profile sensitivity (Methods S8), one-cell tolerance. Promotion and loss are % of DP spans. "Grade loss": provenance read from display grade; "Recorded loss": recorded endpoint provenance. EC-DP promotion and loss were 0 in every row. The last two columns are % of measured cells drawn below solid and of interpolated cells drawn solid. Span counts differ by profile because display-grade runs depend on support.

Profile	Site	Spans	Promotion	Grade loss	Recorded loss	*M* below solid	*I* solid
Reference	White Sands	82	24.4	93.9	0.0	34.9	0.0
	Estonia Tava	172	6.4	100.0	7.6	100.0	0.0
	StREAM	54	22.2	100.0	1.9	94.3	0.0
	Marsh Island	186	4.3	98.4	0.0	39.4	0.0
Exponential	White Sands	73	20.5	13.7	1.4	1.1	0.0
	Estonia Tava	154	1.3	100.0	8.4	87.5	0.0
	StREAM	45	6.7	100.0	4.4	63.0	0.0
	Marsh Island	193	1.0	96.4	0.0	4.6	0.0
Kriging	White Sands	70	14.3	12.9	0.0	0.0	0.1
	Estonia Tava	175	17.1	72.6	8.6	0.0	34.5
	StREAM	59	35.6	86.4	3.4	0.0	21.0
	Marsh Island	192	1.0	96.4	0.0	0.0	0.0
Std. error	White Sands	70	17.1	14.3	0.0	0.0	3.4
	Estonia Tava	157	10.2	75.8	8.9	0.3	98.7
	StREAM	60	31.7	90.0	3.3	2.4	93.0
	Marsh Island	191	2.6	88.0	0.0	0.0	7.6

Table S8. Third-party toolchains (Methods S9): contour lines, simplified spans, vertex reduction, endpoint promotion and provenance loss (counts), and cells whose return count differs from the production binning; every tool's counts are reproduced exactly by its binning rule (Methods S9).

Site	Chain	Lines	Spans	Reduction	Promoted	Prov. loss	Count diff.
White Sands	GDAL	19	89	73.7%	15	0	118
	GRASS	18	86	73.5%	12	0	118
	SAGA	9	62	75.8%	10	0	164
	WhiteboxTools	11	82	91.6%	19	1	118
	lidR	25	78	77.3%	11	0	112
	SciPy/scikit-image	18	64	78.1%	11	0	4
Estonia Tava	GDAL	60	270	55.5%	3	13	245
	GRASS	68	226	57.6%	5	8	245
	SAGA	105	373	51.2%	3	6	216
	WhiteboxTools	36	145	88.8%	5	16	245
	lidR	48	134	71.8%	4	14	240
	SciPy/scikit-image	56	157	68.3%	3	14	4
StREAM	GDAL	14	58	70.1%	2	3	0
	GRASS	16	47	73.0%	2	2	0
	SAGA	14	46	72.7%	6	2	92
	WhiteboxTools	11	57	89.9%	4	1	0
	lidR	13	29	79.9%	3	3	2
	SciPy/scikit-image	13	35	78.8%	2	2	0
Marsh Island	GDAL	18	55	82.4%	8	1	284
	GRASS	16	157	91.9%	46	2	284
	SAGA	17	133	91.3%	28	4	7
	WhiteboxTools	34	155	94.2%	14	2	284
	lidR	51	180	93.5%	40	7	284
	SciPy/scikit-image	26	113	95.4%	38	4	0

Table S9. Attribute query on exported records (Methods S10). Lengths in metres. "Reference": source-edge length that passes the test. "False acc.": failing length inside accepted spans; "Over-rej.": passing length inside rejected spans.

			DP, endpoint records			EC-DP		
Site	t	Reference	Accepted	False acc.	Over-rej.	Accepted	False acc.	Over-rej.
White Sands	66	52.13	40.49	13.77	25.41	6.16	0.00	45.97
White Sands	33	237.46	233.68	9.39	13.17	215.84	0.00	21.61
Estonia Tava	66	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Estonia Tava	33	48.03	41.65	12.33	18.71	22.31	0.00	25.73
StREAM	66	0.00	0.00	0.00	0.00	0.00	0.00	0.00
StREAM	33	15.05	12.57	6.38	8.86	1.69	0.00	13.36
Marsh Island	66	7.42	1.23	0.00	6.19	1.23	0.00	6.19
Marsh Island	33	10.53	2.80	0.27	8.00	1.73	0.00	8.79

Table S10. Conformance and adversarial results (main-text Sections 5.1–5.2).

Check	Result
Measured support, $n = 1 \ldots 40$	Bounded, monotone; range 0.025–0.930
Single-return edge case	$C_M(1; 1, 3) = 0.25$
Interpolation distance sequence	0.50, 0.333, 0.25, 0.20
Eight-direction finite state	256 patterns; 64 one-sided; 0 mismatches
Independent Python check of the evidence rules	14/14 checks passed
Applicability, scope, lineage	6/6 passed in both implementations
EC-DP evidence propagation	Arc payload forwarded; unavailability kept, scope intersected
Python DP/EC-DP differential	2,500/2,500 exact; max support difference 0
Endpoint-only promotion, $\varepsilon = 0.02$	14 687/26 591 spans (55.2%)
Endpoint-only promotion, $\varepsilon = 0.05$	9 830/13 004 spans (75.6%)
Endpoint-only promotion, $\varepsilon = 0.10$	7,678/9,469 spans (81.1%)
Endpoint-only promotion, $\varepsilon = 0.20$	4,650/5,459 spans (85.2%)
Endpoint-only promotion, $\varepsilon = 0.40$	2,000/2,000 spans (100%)
EC-DP promotion	0 at every tolerance; 0 geometry mismatches, including across provenance transitions
Unsupported-gap bridge	baseline 970/1,000; EC-DP 0/1,000
Closed-ring extension	32 reindexings; 1 invariant output; 0 promotions

Table S11. Well-supported contour length (m) that each record states for its DP spans, against the true source length at or above the threshold (Note N4). Endpoint: ordinary DP; minimum: EC-DP; profile: fraction times span length. Rows with no length at the threshold are omitted.

Site	Channel, threshold	True	Endpoint	Minimum	Profile
White Sands	Measured ≥ 66	52.13	43.56	6.16	52.13
	Measured ≥ 33	241.52	248.65	227.74	241.52
	Interpolation ≥ 33	5.10	3.57	3.57	5.10
Estonia Tava	Measured ≥ 33	74.40	48.76	28.49	74.40
	Interpolation ≥ 66	0.00	15.38	0.00	0.00
	Interpolation ≥ 33	201.99	201.99	201.99	201.99
StREAM	Measured ≥ 33	43.72	38.86	11.30	43.72
	Interpolation ≥ 66	0.00	5.26	0.00	0.00
	Interpolation ≥ 33	88.50	94.25	79.00	88.50
Marsh Island	Measured ≥ 66	13.38	10.81	8.10	13.38
	Measured ≥ 33	19.12	19.11	15.34	19.12
	Interpolation ≥ 33	22.03	14.59	14.59	22.03

Table S12. Evidence attributes stored on each contour feature in ten national products (Methods S14). (a) source or generation type; (b) quality flag; (c) accuracy value; (d) link to the source dataset. "Part": the field exists but answers the question only in part. Field names are as published.

Product (agency)	(a)	(b)	(c)	(d)	Fields that answer (a)–(d)
Elevation Contours, large scale (USGS, United States)	No	No	No	Yes	`source_featureid`, `source_datasetid`, `source_datadesc`, `source_originator`
OS Terrain 50 and OS Terrain 5 (Ordnance Survey, Great Britain)	No	No	No	No	None; height and line type only
CanVec contour, catalogue 1.0.2 (Natural Resources Canada)	Yes	Part	Yes	Part	Contour generation (collected, derived); geometric representation qualifier (definite, approximate); altimetric and planimetric accuracy; provider
NTDB contours 100K and 250K (Geoscience Australia)	Yes	No	Yes	Yes	Type "Interpolated Contour"; `FEATURESOURCE`; `PLANIMETRICACCURACY`, `ELEVATIONACCURACY`; `ATTRIBUTESOURCE`
Topo50 contour (LINZ, New Zealand)	No	Part	No	No	`contour_definition` = indefinite
FKB-Høydekurve 5.0.3 (Kartverket, Norway)	Yes	Yes	Yes	Part	`datakilde`, `datafangstmetode`; `sikkerTerrenggjengivelse` (yes/no); `nøyaktighet`, `nøyaktighetHøyde`
N50 Kartdata (Kartverket, Norway)	Part	No	Yes	No	`MåLEMETODE`; `NØYAKTIGHET`
Topografi 50 (Lantmäteriet, Sweden)	No	No	No	No	None; height, type and index flag only
Maastotietokanta contour (National Land Survey of Finland)	Part	No	Yes	Part	`Korkeustarkkuus` (codes KM10, KM25 when height is from an elevation model); `Sijaintitarkkuus`; `Aineistolähde`
Fundamental Geospatial Data contour (GSI, Japan)	No	No	No	Yes	`orgMDId`, `orgGILvl`

Sources, in row order: https://carto.nationalmap.gov/arcgis/rest/services/contours/MapServer; https://docs.os.uk/os-downloads/products/land-and-terrain-portfolio/os-terrain-50/os-terrain-50-technical-specification/gml-feature-types; https://download-telecharger.services.geo.ca/pub/nrcan_rncan/vector/canvec/doc/archive/CanVec_feature_catalogue_v1_0_2.html; https://ga.gov.au/mapspecs/topographic/v6/appendixA_files/Relief.html; https://docs.topo.linz.govt.nz/data-dictionary/tdd-class-contour.html; https://dokument.geonorge.no/produktspesifikasjoner/fkb-h%C3%B8ydekurve/Versjon%205.0.3/fkb-h%C3%B8ydekurve-503.pdf; https://register.geonorge.no/data/documents/Produktspesifikasjoner_N50%20Kartdata_v14_produktspesifikasjon-kartverket-n50kartdata-20170401_.pdf; https://geotorget.lantmateriet.se/dokument/projects/topografi-50-nedladdning-vektor/released/2026.05/Tema-H%C3%B6jd.html; https://www.maanmittauslaitos.fi/sites/maanmittauslaitos.fi/files/old/maastotietokohteet.pdf; https://service.gsi.go.jp/kiban/contents/screen/basismap/documents/FGD_DLFileSpecV5.2.pdf.

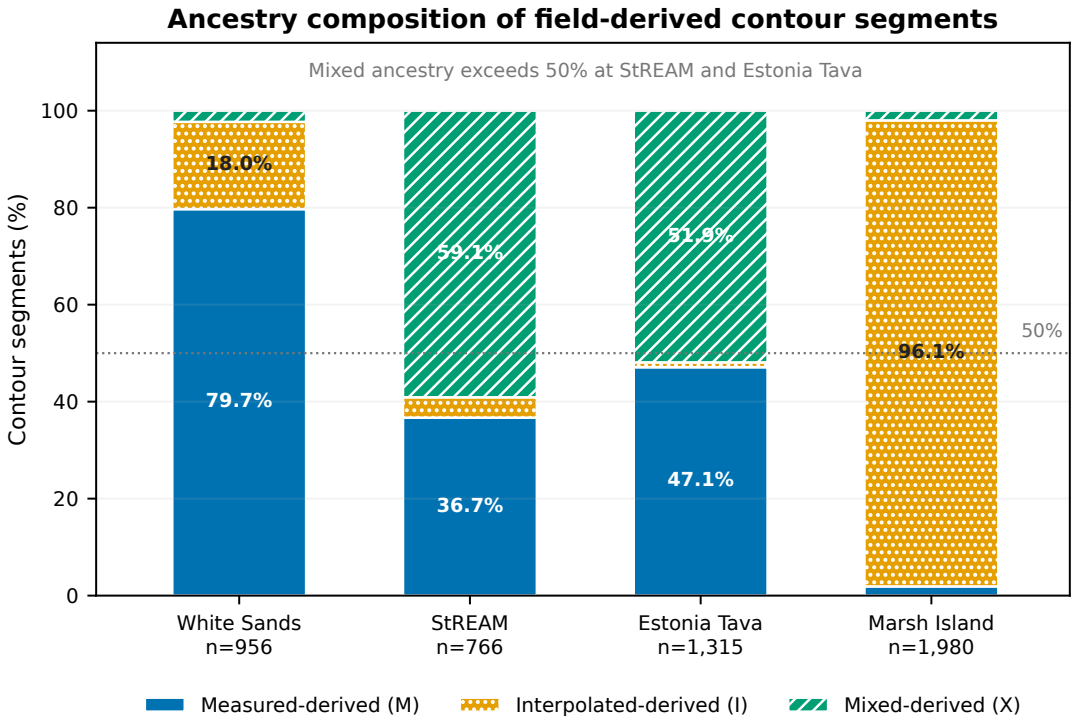


Figure S2. Composition of measured-derived (M), interpolated-derived (I) and mixed (X) ancestry in the field contour segments. Mixed ancestry exceeds 50% in StREAM and Estonia Tava.

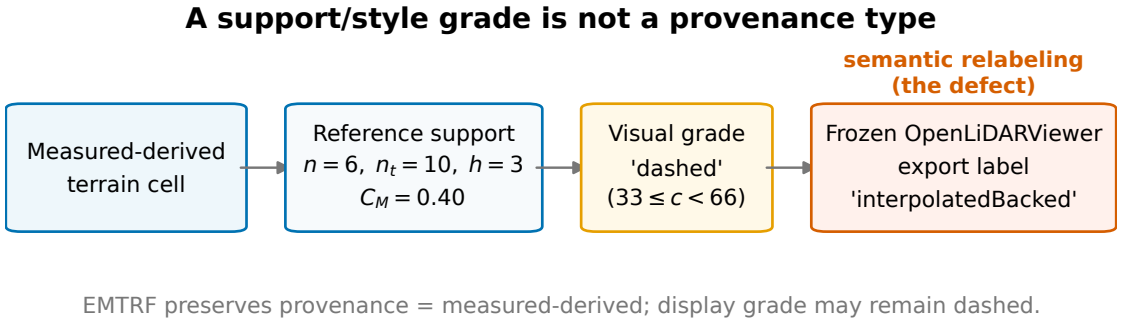


Figure S3. Source-level grade–provenance counterexample (main-text Section 5.6). A measured-derived case has $C_M = 0.40$ ($c = 100\,C_M$ in the figure) while the audited export helper maps the dashed style to `interpolatedBacked`. The case stays dashed under 25/75 and 40/60 thresholds. EMTRF keeps provenance, support and display grade as separate fields.